
General principles and requirements for bodies validating and verifying environmental information

*Principes généraux et exigences pour les organismes de validation et
de vérification de l'information environnementale*





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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT), see www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 207, *Environmental management*, Subcommittee SC 7, *Greenhouse gas management and related activities*, in collaboration with the European Committee for Standardization (CEN) Technical Committee CEN/SS S26, *Environmental management*, in accordance with the Agreement on technical cooperation between ISO and CEN (Vienna Agreement).

This third edition cancels and replaces the second edition (ISO 14065:2013), which has been technically revised. The main changes compared with the previous edition are as follows:

- the Scope has been expanded to include bodies performing validation, verification and agreed upon procedures in all areas of environmental information (not only greenhouse gas);
- it has been aligned with the requirements of ISO/IEC 17029;
- [Annex D](#) has been added for additional requirements applicable to green bonds;
- [Annex E](#) has been added for additional requirements applicable to greenhouse gases;
- [Annex F](#) has been added for additional requirements applicable to non-financial disclosure.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

Introduction

Environmental information is increasingly being used for decision-making by individuals and organizations. Environmental information can be seen in various forms, including but not limited to:

- greenhouse gas statements;
- environmental footprints (e.g. carbon and water);
- environmental performance;
- environmental labelling claims, including environmental product declarations;
- environmental information as part of sustainability reporting;
- calculations associated with the valuation of environmental resources;
- environmental information related to “green bonds”, “climate finance” and other financial instruments.

Users of environmental information want to know whether it is accurate and reliable. They seek assurance on statements of historical information, and validation that forecasted information is based on reasonable assumptions and methods. This document identifies principles and defines requirements for validation and verification bodies that meet these needs.

Requirements for validation/verification bodies in this document include:

- general requirements (see [Clause 5](#), including legal, impartiality, liability, and contractual matters);
- structural requirements (see [Clause 6](#), including organizational structure and operational control);
- resource requirements (see [Clause 7](#), including personnel and competency management);
- programme requirements (see [Clause 8](#));
- process requirements (see [Clause 9](#), including process steps such as pre-engagement, engagement, execution, review, and issuance of opinions and records management);
- information requirements (see [Clause 10](#), including communication and confidentiality);
- management systems requirements (see [Clause 11](#), including the internal management system of the body, internal audits, management review and corrective actions).

This document is a sector application of ISO/IEC 17029:2019. It references the requirements of ISO/IEC 17029 and also includes specific requirements related to bodies that validate or verify environmental information.

Bodies operating in accordance with this document can be first-party, second-party or third-party bodies. Bodies can provide validation only, verification only, or provide both validation and verification, and perform agreed-upon procedures (AUP).

This document provides programme owners, regulators and accreditation bodies with a basis for assessing and recognizing the competence of validation and verification bodies. It can also be used in other ways, such as in peer assessment within groups of validation/verification bodies or between such groups.

General principles and requirements for bodies validating and verifying environmental information

1 Scope

This document specifies principles and requirements for bodies performing validation and verification of environmental information statements.

Any programme requirements related to bodies are additional to the requirements of this document.

This document is a sector application of ISO/IEC 17029:2019, which contains general principles and requirements for the competence, consistent operation and impartiality of bodies performing validation/verification as conformity assessment activities.

This document includes sector-specific requirements in addition to the requirements of ISO/IEC 17029:2019.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 14030-4,¹⁾ *Environmental performance evaluation — Green debt instruments — Part 4: Verification programme requirements*

ISO 14064-3, *Greenhouse gases — Part 3: Specification with guidance for the verification and validation of greenhouse gas statements*

ISO 14066, *Greenhouse gases — Competence requirements for greenhouse gas validation teams and verification teams*

ISO 14097,²⁾ *Framework including principles and requirements for assessing and reporting investments and financing activities related to climate change*

ISO/IEC 17000, *Conformity assessment — Vocabulary and general principles*

ISO/IEC 17029:2019, *Conformity assessment — General principles and requirements for validation and verification bodies*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO/IEC 17000, ISO/IEC 17029:2019 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <http://www.electropedia.org/>

1) Under preparation. Stage at the time of publication: ISO/DIS 14030-4:2020.

2) Under preparation. Stage at the time of publication: ISO/DIS 14097:2020.

3.1 Terms related to environmental information

3.1.1

environment

surroundings in which an *organization* (3.2.2) operates, including air, water, land, natural resources, flora, fauna, humans and their interrelationships

Note 1 to entry: Surroundings can extend from within an organization to the local, regional and global system.

Note 2 to entry: Surroundings can be described in terms of biodiversity, ecosystems, climate or other characteristics.

[SOURCE: ISO 14001:2015, 3.2.1]

3.1.2

environmental aspect

element of an *organization's* (3.2.2) activities or products or services that interacts or can interact with the *environment* (3.1.1)

[SOURCE: ISO 14001:2015, 3.2.2, modified — The notes to entry have been deleted.]

3.1.3

environmental performance

measurable results related to the management of *environmental aspects* (3.1.2)

[SOURCE: ISO 14001:2015, 3.4.11, modified — “measurable results” have replaced the “performance”.]

3.1.4

environmental information

subject matter of a qualitative or quantitative nature that is related to environmental conditions or *environmental performance* (3.1.3)

Note 1 to entry: Environmental information can include statements and claims regarding greenhouse gas emissions, removals, emission reductions or removal enhancements of an *organization* (3.2.2), project (e.g. see ISO 14064-1 and ISO 14064-2), environmental footprints (e.g. see ISO 14067 for carbon footprints of a product, ISO 14046 for water footprints and ISO 14044 for life cycle assessment information) or environmental reports (e.g. see ISO 14016).

Note 2 to entry: ISO 14033 defines and specifies terms and procedures to establish reviewable and comparable quantitative environmental information.

3.1.5

environmental information statement

declaration of *environmental information* (3.1.4)

Note 1 to entry: The environmental information statement can represent a point in time or can cover a period of time.

Note 2 to entry: The environmental information statement provided by the *responsible party* (3.2.3) should be clearly identifiable and capable of consistent evaluation or measurement against suitable *criteria* (3.3.20) by a *verifier* (3.3.5) or *validator* (3.3.6).

Note 3 to entry: The environmental information statement can be provided in: a report; a declaration; an economic, financial or monetary valuation; an environmental product declaration; a life cycle assessment report; a climate change vulnerability or adaptation evaluation; a project plan; a label or logo.

Note 4 to entry: The term “environmental information statement” corresponds to the term “claim” used in ISO/IEC 17029:2019, 3.1.

3.1.6

environmental information programme

rules and procedures for providing an *environmental information statement* (3.1.5)

Note 1 to entry: Environmental information programmes can be carried out at international, regional, national or sub-national levels.

Note 2 to entry: A programme can also be called a scheme.

Note 3 to entry: Greenhouse gas emissions and reductions, greenhouse gas inventories, carbon and water footprints, and the *environmental information* (3.1.4) in sustainability reports are examples of subjects that may be verified in accordance with an environmental information programme.

Note 4 to entry: An environmental information programme may include requirements for *validation* (3.3.16) or *verification* (3.3.15).

3.2 Terms related to personnel and organization

3.2.1

client

organization (3.2.2) or person requesting *validation* (3.3.16) or *verification* (3.3.15)

Note 1 to entry: The client could be the *responsible party* (3.2.3), *programme owner* (3.3.3), *intended user* (3.2.4) or other interested party.

[SOURCE: ISO/IEC 17029:2019, 3.13, modified — Note 1 to entry has been added.]

3.2.2

organization

person or group of people that has its own functions with responsibilities, authorities and relationships to achieve its objectives

[SOURCE: ISO 14001:2015, 3.1.4, modified — Note 1 to entry has been deleted.]

3.2.3

responsible party

person or persons responsible for the provision of the *environmental information statement* (3.1.5) and the supporting information

Note 1 to entry: The responsible party can be either individuals or authorized representatives of an *organization* (3.2.2) or project and can be the party who engages the *verifier* (3.3.5) or *validator* (3.3.6).

Note 2 to entry: The responsible party may be the *client* (3.2.1).

[SOURCE: ISO 14064-3:2019, 3.2.3, modified — The references to greenhouse gas and product have been deleted. Note 2 to entry has been added.]

3.2.4

intended user

individual or *organization* (3.2.2) identified by those reporting *environmental information* (3.1.4) as being the one who relies on that environmental information to make decisions

Note 1 to entry: The intended user could be the *client* (3.2.1), the *responsible party* (3.2.3), *programme owners* (3.3.3), regulators, the financial community, the general public or other interested parties, such as local communities, governmental or non-governmental organizations.

[SOURCE: ISO 14064-3:2019, 3.2.4, modified — The references to greenhouse gas have been deleted. Note 1 to entry has been revised.]

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3.2.5

technical expert

person who provides specific knowledge or expertise to the *validation/verification team* (3.3.7)

Note 1 to entry: Specific knowledge or expertise is that which relates to the *organization* (3.2.2), the process or activity associated with the subject to be verified or validated, finance, local regulations, language or culture.

Note 2 to entry: A technical expert does not act as a *verifier* (3.3.5) or *validator* (3.3.6) in the validation/verification team.

[SOURCE: ISO 19011:2018, 3.16, modified — “validation/verification team” has replaced “audit team” in the definition and Note 2 to entry. Note 1 to entry has been revised.]

3.2.6

competence

ability to apply knowledge and skills to achieve intended results

[SOURCE: ISO 14066:2011, 3.1.4, modified — The notes to entry have been deleted.]

3.2.7

nonconformity

non-fulfilment of a requirement

Note 1 to entry: Some programmes can require that nonconformities raised against legal requirements related to the programme are raised as noncompliances.

[SOURCE: ISO 14064-3:2019, 3.6.19, modified — Note 1 to entry has been added.]

3.3 Terms related to validation and verification

3.3.1

validation programme

rules, procedures and management for carrying out *validation* (3.3.16) activities in a specific sector or field

Note 1 to entry: Validation programmes may be operated at international, regional, national, sub-national, sector-specific or organizational levels.

Note 2 to entry: A programme can also be called a “scheme”.

Note 3 to entry: A set of standards able to cover all the requirements of this document can serve as a programme.

Note 4 to entry: A validation programme can be as simple as the letter of *engagement* (3.3.13) between the *validation body* (3.3.26) and its *client* (3.2.1), or refer to a formal programme document which has a set of rules that can have varying degrees of specification and complexity.

[SOURCE: ISO/IEC 17029:2019, 3.8, modified — In the definition, “or field” has been added. In Note 1 to entry, “organizational” has been added. Note 4 to entry has been added.]

3.3.2

verification programme

rules, procedures and management for carrying out *verification* (3.3.15) activities in a specific sector or field

Note 1 to entry: Verification programmes may be operated at international, regional, national, sub-national, sector-specific or organizational levels.

Note 2 to entry: A programme can also be called a “scheme”.

Note 3 to entry: A set of standards able to cover all the requirements of this document can serve as a programme.

Note 4 to entry: A verification programme can be as simple as the letter of *engagement* (3.3.13) between the *verification body* (3.3.27) and its *client* (3.2.1), or refer to a formal programme document which has a set of rules that can have varying degrees of specification and complexity.

[SOURCE: ISO/IEC 17029:2019, 3.9, modified — “or field” has been added to the definition. Note 4 to entry has been added.]

3.3.3

programme owner

person or *organization* (3.2.2) responsible for developing and maintaining an *environmental information programme* (3.1.6), a *validation programme* (3.3.1) or a *verification programme* (3.3.2)

Note 1 to entry: The programme owner can be the body itself, a governmental authority, a trade association, a group of validation/verification bodies, an external programme owner or others.

Note 2 to entry: The programme owner can be the scheme owner.

[SOURCE: ISO/IEC 17029:2019, 3.10, modified — “environmental information programme” has replaced “specific” added. Note 2 to entry has been added.]

3.3.4

team leader

person who manages the *validation/verification team* (3.3.7)

[SOURCE: ISO 14066:2011, 3.1.2]

3.3.5

verifier

competent and impartial person with responsibility for performing and reporting on a *verification* (3.3.15)

[SOURCE: ISO 14064-3:2019, 3.2.6]

3.3.6

validator

competent and impartial person with responsibility for performing and reporting on a *validation* (3.3.16)

[SOURCE: ISO 14064-3:2019, 3.2.7]

3.3.7

validation/verification team

one or more persons conducting *validation* (3.3.16)/*verification* (3.3.15) activities

Note 1 to entry: One person of the validation/verification team is appointed as the team leader.

Note 2 to entry: The validation/verification team may be accompanied by validators-in-training or verifiers-in-training.

[SOURCE: ISO 14064-3:2019, 3.2.8, modified — Note 2 to entry has been added.]

3.3.8

independent reviewer

competent person, who is not a member of the *validation/verification team* (3.3.7), who reviews the *verification* (3.3.15) or *validation* (3.3.16) activities and conclusions

[SOURCE: ISO 14064-3:2019, 3.2.9]

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3.3.9

impartiality

presence of objectivity

Note 1 to entry: Objectivity means that conflicts of interest do not exist, or are resolved so as not to adversely influence the activities of the body.

Note 2 to entry: Other terms that are useful in conveying the element of impartiality include “independence”, “freedom from conflicts of interest”, “freedom from bias”, “lack of prejudice”, “neutrality”, “fairness”, “open-mindedness”, “even-handedness”, “detachment” “balance”.

[SOURCE: ISO/IEC 17021-1:2015, 3.2, modified — “certification” has been deleted before “body” in Note 1 to entry.]

3.3.10

consultancy

provision of specific expertise on the subject matter that supports the preparation of an *environmental information statement* (3.1.5)

Note 1 to entry: Arranging training and participating as a trainer is not considered consultancy, provided that, where the course relates to *validation* (3.3.16) and *verification* (3.3.15) or the environmental information statement being validated or verified, it is confined to the provision of generic information, i.e. the trainer should not provide client-specific advice or solutions.

3.3.11

appeal

request to the body for reconsideration of a decision it has made with respect to the issuance of a *validation* (3.3.25) or *verification opinion* (3.3.23)

3.3.12

complaint

expression of dissatisfaction, other than *appeal* (3.3.11), by any person or *organization* (3.2.2) to a body, relating to the activities of that body, where a response is expected

[SOURCE: ISO/IEC 17000:2020, 8.7, modified — “body” has replaced “conformity assessment body or an accreditation body”.]

3.3.13

engagement

arrangement between the *validation* (3.3.26) or *verification body* (3.3.27) and its *client* (3.2.1) with the terms to perform services, usually specified in the form of a contract

Note 1 to entry: The word “engagement” is also sometimes used to refer to the activities performed under an engagement, such as a *validation* (3.3.16) or a *verification* (3.3.15), or an agreement to perform *agreed-upon procedures* (3.3.17).

3.3.14

assurance

confidence in an *environmental information statement* (3.1.5) that is historical in nature

3.3.15

environmental information verification verification

process for evaluating an *environmental information statement* (3.1.5) based on historical data and information to determine whether the statement is materially correct and conforms to *criteria* (3.3.20)

Note 1 to entry: Verification activities performed that do not lead to the expression of an opinion are called *agreed-upon procedures* (3.3.17).

Note 2 to entry: The term “environmental information verification” is shortened to “verification” in this document to reduce sentence complexity and aid understanding.

3.3.16 environmental information validation validation

process for evaluating the reasonableness of the assumptions, limitations and methods that support an *environmental information statement* (3.1.5) about the outcome of future activities

Note 1 to entry: The term “environmental information validation” is shortened to “validation” in this document to reduce sentence complexity and aid understanding.

3.3.17 agreed-upon procedures AUP

engagement (3.3.13) that reports on the results of *verification* (3.3.15) activities and does not provide an *opinion* (3.3.23)

Note 1 to entry: Agreed-upon procedures do not provide *assurance* (3.3.14).

[SOURCE: ISO 14064-3:2019, 3.6.4, modified — Note 1 to entry has been added.]

3.3.18 level of assurance degree of confidence in the *environmental information statement* (3.1.5)

Note 1 to entry: *Assurance* (3.3.14) is provided on historical information.

[SOURCE: ISO 14064-3:2019, 3.6.5, modified — “environmental information statement” has replaced “GHG statement”. Note 1 to entry has been added.]

3.3.19 materiality concept that individual *misstatements* (3.3.21) or the aggregation of misstatements could influence the *intended users’* (3.2.4) decisions

[SOURCE: ISO 14064-3:2019, 3.6.9]

3.3.20 criteria policies, procedures or requirements used as a reference against which the *environmental information statement* (3.1.5) is compared

Note 1 to entry: Criteria may be established by governments, regulators, *environmental information programmes* (3.1.6), voluntary reporting initiatives, standards, codes of practice, or internal procedures.

Note 2 to entry: “Criteria” is used in place of “specified requirements” used in ISO/IEC 17029.

[SOURCE: ISO 14064-3:2019, 3.6.10, modified — “environmental information statement” has replaced “GHG statement”. Notes 1 and 2 to entry have been added.]

3.3.21 misstatement error, omission, misreporting or misrepresentation in the *environmental information statement* (3.1.5)

Note 1 to entry: Misstatement can be qualitative or quantitative.

[SOURCE: ISO 14064-3:2019, 3.6.15, modified — “environmental information statement” has replaced “GHG statement”. Note 1 to entry has been added.]

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3.3.22

material misstatement

individual *misstatement* (3.3.21) or the aggregate of actual misstatements in the *environmental information statement* (3.1.5) that could affect the decisions of the *intended users* (3.2.4)

[SOURCE: ISO 14064-3:2019, 3.6.17, modified — “environmental information statement” has replaced “GHG statement”.]

3.3.23

verification opinion

formal written declaration to the *intended user* (3.2.4) that provides confidence that the *environmental information statement* (3.1.5) is materially correct and conforms with the *criteria* (3.3.20)

Note 1 to entry: The term “verification opinion” is a type of “verification statement” in ISO/IEC 17029:2019, 3.7.

[SOURCE: ISO 14064-3:2019, 3.6.18, modified — In the term, “/validation” has been deleted. In the definition, “that the environmental information statement is materially correct” has replaced “on the GHG statement in the responsible party’s GHG report”. Note 1 to entry has been added.]

3.3.24

report of factual findings

documented output of *agreed-upon procedures* (3.3.17)

Note 1 to entry: The term “report of factual findings” is a type of “verification statement” in ISO/IEC 17029:2019, 3.7.

3.3.25

validation opinion

formal written declaration to the *intended user* (3.2.4) on the reasonableness of the assumptions, methods and limitations used to develop forecasts and projections contained in the *environmental information statement* (3.1.5)

Note 1 to entry: The term “validation opinion” is a type of “validation statement” in ISO/IEC 17029:2019, 3.6.

Note 2 to entry: Reasonableness of the assumptions, methods and limitations includes consideration of conformity to applicable *criteria* (3.3.20).

[SOURCE: ISO 14064-3:2019, 3.6.18, modified— In the term, “verification/” has been deleted. In the definition, “on the reasonableness of the assumptions, methods and limitations used to develop forecasts and projections contained in the environmental information statement” has replaced “on the GHG statement in the responsible party’s GHG report and confirms conformity to the criteria”. Notes 1 and 2 to entry have been added.]

3.3.26

validation body

body that performs *validation* (3.3.16)

Note 1 to entry: A validation body can be an *organization* (3.2.2), or part of an organization.

Note 2 to entry: The term “the validation/verification body” is shortened to “the body” in this document to reduce sentence complexity and aid understanding.

[SOURCE: ISO/IEC 17029:2019, 3.4, modified — Note 2 to entry has been added.]

3.3.27

verification body

body that performs *verification* (3.3.15)

Note 1 to entry: A verification body can be an *organization* (3.2.2), or part of an organization.

Note 2 to entry: The term “the validation/verification body” is shortened to “the body” in this document to reduce sentence complexity and aid understanding.

[SOURCE: ISO/IEC 17029:2019, 3.5, modified — Note 2 to entry has been added.]

4 Principles

4.1 General

ISO/IEC 17029:2019, 4.1, shall be followed.

NOTE The reference to “specified requirements” in ISO/IEC 17029 means “criteria” in this document.

4.2 Principles for the validation/verification process

ISO/IEC 17029:2019, 4.2, shall be followed.

4.3 Principles for validation/verification bodies

ISO/IEC 17029:2019, 4.3, shall be followed.

4.4 Conservativeness

When the body assesses comparable alternatives, preference is given to the alternative that is cautiously moderate.

4.5 Professional scepticism

Attitude based on recognition of the potential circumstances that can cause material misstatements in an environmental information statement.

NOTE ISO 14066:2011, Annex A, provides guidance on evidence and the application of professional scepticism. This annex applies equally to the validation and verification of all environmental information.

5 General requirements

5.1 Legal entity

ISO/IEC 17029:2019, 5.1, shall be followed.

The body shall have a documented description of its legal status including, if applicable, the names of its owners and, if different, the names of the persons who control it.

5.2 Responsibility for validation/verification statements

ISO/IEC 17029:2019, 5.2, shall be followed.

NOTE The reference to “validation/verification statements” in ISO/IEC 17029 means “validation/verification opinions” in this document.

The body shall be responsible for the activities that it performs in AUP engagements and for the reports of factual findings that it issues as a result of the application of the procedures.

5.3 Management of impartiality

ISO/IEC 17029:2019, 5.3, shall be followed.

The body shall ensure, through a mechanism independent of its operations, that impartiality is being achieved.

5.4 Liability

ISO/IEC 17029:2019, 5.4, shall be followed.

6 Structural requirements

6.1 Organizational structure and top management

ISO/IEC 17029:2019, 6.1, shall be followed.

6.2 Operational control

ISO/IEC 17029:2019, 6.2, shall be followed.

7 Resource requirements

7.1 General

ISO/IEC 17029:2019, 7.1, shall be followed.

7.2 Personnel

ISO/IEC 17029:2019, 7.2, shall be followed.

For ISO/IEC 17029:2019, 7.2.4, note that verifiers and validators demonstrate compliance with ethical requirements by adhering to the principles included in [Clause 4](#).

For ISO/IEC 17029:2019, 7.2.5, the period specified shall not be less than two years.

7.3 Management process for the competence of personnel

7.3.1 ISO/IEC 17029:2019, 7.3, shall be followed.

7.3.2 In addition to having the process required by ISO/IEC 17029:2019, 7.3.1, the body shall establish, implement and maintain a process for:

- a) defining required competencies for each programme and sector in which it operates;
- b) ensuring that verifiers, validators, technical experts and reviewers have appropriate competencies;
- c) ensuring that there is access to relevant internal or external expertise for advice on specific matters relating to the environmental information programme, validation/verification activities, sectors or areas within the scope of their work.

The additional requirements and competencies for personnel given in [Annexes D, E and F](#) shall be followed as applicable.

7.3.3 Regarding ISO/IEC 17029:2019, 7.3.3, note that performance monitoring shall be periodic. Monitoring techniques may include annual performance reviews, review of the reports, on the job monitoring and interviews. The monitoring techniques used shall be in proportion with the impact of the performance on the outcome of the validation/verification.

7.3.4 The body shall establish competent validation/verification teams and shall provide appropriate management and support services. If one individual fulfils all the requirements for a validation/verification team, then that person may be considered as a validation/verification team.

7.3.5 The validation/verification team shall have the ability to apply detailed knowledge of the applicable programme, including its:

- a) eligibility requirements;
- b) implementation in different jurisdictions, as applicable;
- c) validation or verification requirements and guidelines.

7.3.6 The validation/verification team shall have sufficient technical expertise to evaluate:

- a) relevant activities and technologies;
- b) quantification, monitoring and reporting, including relevant technical and sector issues

7.3.7 The validation/verification team shall have data and information auditing expertise to evaluate the environmental information statement, including the ability:

- a) to evaluate the information system to determine whether the responsible party has effectively identified, collected, analysed and reported on relevant environmental information, and has systematically taken corrective actions to address any misstatements and nonconformities;
- b) to design an evidence-gathering plan;
- c) to analyse risks associated with the use of data and data systems;
- d) to identify failures in data and data systems;
- e) to evaluate the impact of the various streams of data on the materiality of the environmental information statement.

7.3.8 The validation/verification team shall be able to communicate effectively in appropriate languages on matters relevant to the validation or verification.

7.3.9 The validation/verification team leader shall have:

- a) sufficient knowledge and expertise of the competencies detailed in [7.3.1](#) to [7.3.5](#) to manage the validation/verification team in order to meet the validation or verification objectives;
- b) the demonstrated ability to perform a validation or verification;
- c) the demonstrated ability to manage audit teams.

7.4 Outsourcing

ISO/IEC 17029:2019, 7.4, shall be followed.

For ISO/IEC 17029:2019, 7.4 b), note that “engagement activities” refers to the process by which an agreement between the client and the body is concluded.

8 Validation/verification programme

ISO/IEC 17029:2019, Clause 8, shall be followed.

9 Process requirements

9.1 General

ISO/IEC 17029:2019, 9.1, shall be followed.

9.2 Pre-engagement

ISO/IEC 17029:2019, 9.2, shall be followed.

NOTE A statement to be validated and verified is equivalent to a proposed claim in ISO/IEC 17029.

In addition to the requirements given in ISO/IEC 17029:2019, 9.2.2, the validation/verification team shall ensure that the engagement type(s) has(ve) been identified.

Engagement type(s) may include verification, validation, AUP or a combination thereof.

9.3 Engagement

ISO/IEC 17029:2019, 9.3, shall be followed.

In addition to the requirements given in ISO/IEC 17029:2019, 9.3.2, the client shall communicate any facts to the body that can affect the validity of an issued opinion.

9.4 Planning

9.4.1 ISO/IEC 17029:2019, 9.4, shall be followed.

9.4.2 In addition to the planning activities required in ISO/IEC 17029:2019, 9.4.1, the validation/verification team shall:

- a) perform a strategic analysis to understand the nature and complexity related to the environmental information statement and to determine the extent of the validation/verification activities based on the engagement type;
- b) assess the risk of nonconformity to the criteria.

[Figure 1](#) shows the general planning process.

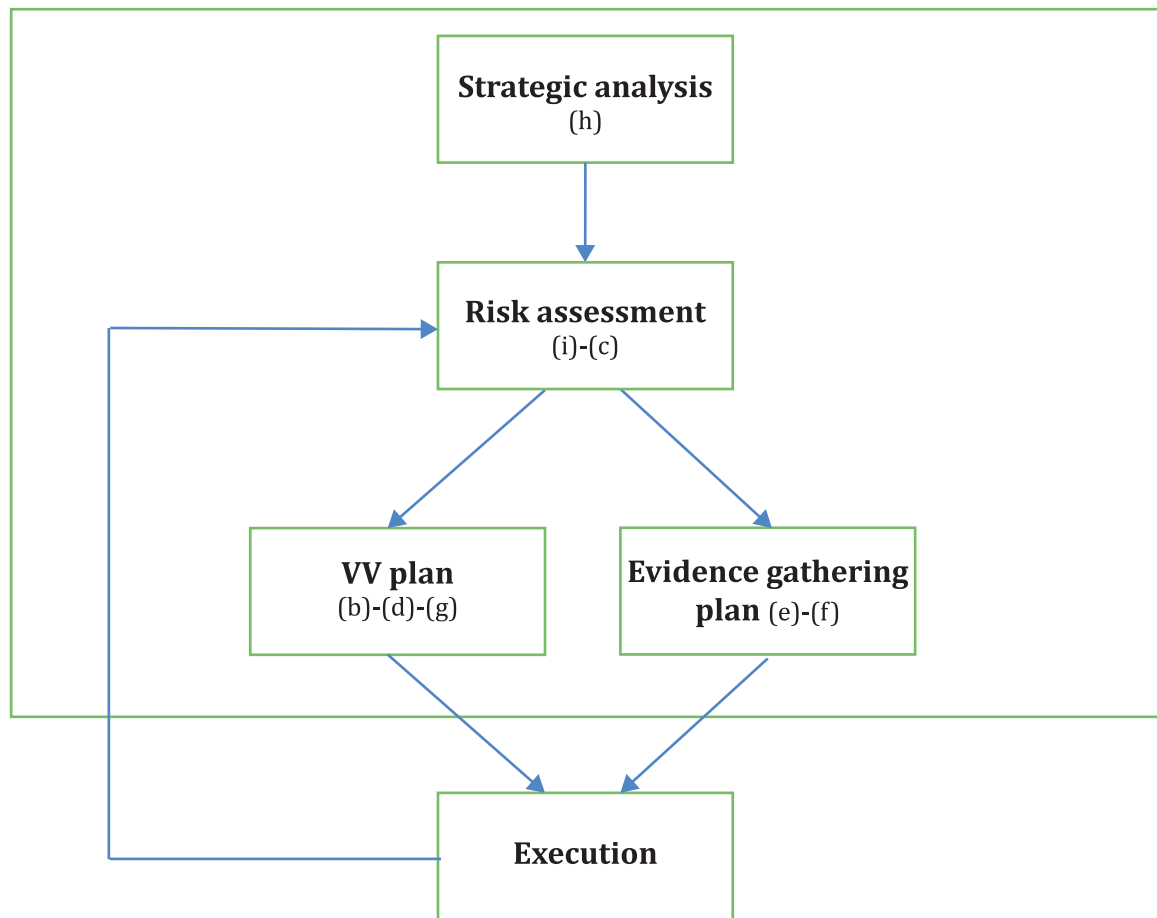


Figure 1 — Steps of a planning process

9.4.3 In addition to the requirements of ISO/IEC 17029:2019, 9.4.2, the validation/verification plan shall include the level of assurance and materiality.

9.4.4 The validation/verification plan and evidence-gathering plan shall be approved by the team leader.

9.4.5 Amendments to the validation/verification plan and evidence-gathering plan shall be approved by the team leader in the following circumstances:

- a) change in scope or timing of validation/verification activities;
- b) change in evidence-gathering procedures;
- c) change in locations and sources of information for evidence-gathering;
- d) when the validation/verification process identifies new risks or concerns that could lead to material misstatements or nonconformities.

9.5 Validation/verification execution

ISO/IEC 17029:2019, 9.5, shall be followed.

NOTE Guidance for sufficient and objective evidence is provided in ISO 14066:2011, Annex A.

9.6 Review

9.6.1 ISO/IEC 17029:2019, 9.6, shall be followed.

9.6.2 In addition to the requirements in ISO/IEC 17029:2019, 9.6.2, the review shall be carried out by persons who have not been involved in the planning (see [9.4](#)) and are not part of the validation/verification team.

NOTE As long as personnel conducting a review have not participated in validation or verification activities under the direction of the team leader, they are not considered members of the validation/verification team.

9.6.3 For ISO/IEC 17029:2019, 9.6.3 c), note that “significant findings” are misstatements and nonconformities identified by the validation/verification team that could affect the opinion.

9.6.4 In addition to the requirements of ISO/IEC 17029:2019, 9.6.3, the review shall confirm:

- a) the competencies of validation/verification team members for the activities that they conducted;
- b) whether the validation/verification planning has been designed appropriately, including whether the objective, scope and materiality are addressed by:
 - 1) the strategic analysis and risk assessment;
 - 2) the validation/verification plan;
 - 3) the evidence-gathering plan;
- c) significant decisions made by the validation/verification team during the validation/verification;
- d) whether the opinion is appropriately drafted;
- e) whether the environmental information statement is fairly stated and conforms to criteria.

9.6.5 In addition to the requirements of ISO/IEC 17029:2019, 9.6.5, the review may be started at any time during the process before the opinion is issued to allow significant issues identified by the reviewer to be resolved, provided that the independence of the reviewer is maintained, and the activities planned and undertaken by the reviewer(s), including the results, are documented.

NOTE A reviewer is sometimes referred to as an “independent reviewer” as given in ISO 14064-3:2019, 3.2.9.

9.6.6 The review shall be completed before the final opinion, or the report of factual finding for the AUP, is issued.

9.7 Decision and issue of the validation/verification statement

9.7.1 Decision

9.7.1.1 ISO/IEC 17029:2019, 9.7.1, shall be followed.

9.7.1.2 Regarding ISO/IEC 17029:2019, 9.7.1.1, note that the reference to the word “claim” means “environmental information statement” in this document. A claim can be confirmed when the body concludes that the claim is materially correct and conforms with specified criteria.

9.7.1.3 The validation/verification body shall decide whether to confirm an environmental information statement that it has tested using AUP in a mixed engagement. The decision shall be based upon the body’s report of factual findings (see [Annex C](#)).

9.7.1.4 Regarding ISO/IEC 17029:2019, 9.7.1.2, note that the person assigned to make the decision may be the reviewer. The decision shall be made by persons who have not been involved in the validation/verification planning (see [9.4](#)).

9.7.1.5 Regarding ISO/IEC 17029:2019, 9.7.1.3, note that the reference to the word “statement” means “verification opinion or validation opinion” in this document.

In the case of AUP, the decision is issued through a report of factual findings.

Bodies may choose not to issue an opinion when the engagement is terminated prior to completion.

9.7.1.6 If an opinion is issued, the body shall select one type of opinion, such as:

- a) unmodified;
- b) modified;
- c) adverse.

NOTE [Annex A](#) describes types of opinions, including unmodified, modified, adverse and variant terminology.

9.7.1.7 The body may disclaim the issuance of an opinion when it is unable to obtain sufficient and appropriate evidence to come to a conclusion. In this case, the body shall ensure that it has been unable to obtain sufficient appropriate evidence and can conclude that the possible effects on the environmental information statement of undetected material misstatement(s) are material and pervasive (see [Tables A.1](#) and [A.2](#)).

9.7.1.8 At the conclusion of an engagement to verify statements of historical information, the verification body shall issue an opinion, unless it has disclaimed the issuance of an opinion or the engagement type is AUP. An opinion providing assurance to intended users shall be based upon the verification of sufficient and appropriate historical evidence.

NOTE Only unmodified or modified opinions provide assurance to intended users.

9.7.1.9 At the conclusion of an engagement to validate statements about the outcome of future activities, the validation body shall issue an opinion, unless it has disclaimed the issuance of an opinion. A validation opinion on the reasonableness of the assumptions, limitations and methods used to forecast information shall be based upon the evaluation of sufficient and appropriate information.

9.7.2 Issue of the validation/verification statement

ISO/IEC 17029:2019, 9.7.2, shall be followed.

If the environmental information statement includes a mixture of hypothetical, projected and/or historical information, the validation and verification opinion may be included in the same document.

The opinion shall contain:

- identification of the environmental information-related activity (e.g. organization, project or product);
- identification of the responsible party;
- a statement that the environmental information statement is the responsibility of the responsible party;
- identification of the criteria agreed by the responsible party and the body for the development of the environmental information statement;

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- identification of the criteria used by the body to validate or verify the environmental information statement;
- where the environmental information statement includes future predictions, an explanation that the actual result can differ from the estimate because the assumptions upon which the estimate is based can change.

The opinion may contain statements that limit the liability of the body.

A modified opinion shall contain a description of the reason for the modification. If the reason for the modified opinion is quantitative, the body's opinion shall indicate the value of the material misstatement and its effect on the environmental information statement.

An adverse opinion shall include the reason(s) for the adverse opinion.

When disclaiming the issuance of an opinion, the body shall provide an explanation.

9.8 Facts discovered after the issue of the validation/verification statement

ISO/IEC 17029:2019, 9.8, shall be followed.

9.9 Handling of appeals

ISO/IEC 17029:2019, 9.9, shall be followed.

9.10 Handling of complaints

ISO/IEC 17029:2019, 9.10, shall be followed.

9.11 Records

ISO/IEC 17029:2019, 9.11, shall be followed.

10 Information requirements

10.1 Publicly available information

ISO/IEC 17029:2019, 10.1, shall be followed.

Publicly provided information shall include any requirements regarding the use of the body's opinion in its entirety (see [Annex B](#)).

10.2 Other information to be available

10.2.1 ISO/IEC 17029:2019, 10.2, shall be followed.

10.2.2 For ISO/IEC 17029:2019, 10.2.2, note that the status of the validation/verification opinion can be confirmation of the identity of the body that issued the opinion, its date of issuance and, if applicable, the revision date.

10.2.3 In addition to the requirements of ISO/IEC 17029:2019, 10.2.3, the validation/verification team shall provide a detailed description of the validation/verification process.

NOTE The description of the validation/verification process includes how the body considers previous validation/verification results, where appropriate and if available.

10.3 Reference to validation/verification and use of marks

10.3.1 ISO/IEC 17029:2019, 10.3, shall be followed.

NOTE Information on third-party marks of conformity is given in ISO/IEC 17030.

10.3.2 The body shall ensure its agreement requires that the client shall not use the environmental information statement, opinion, report, marks, logos or labels in a manner that could mislead intended users or impair the reputation of the body.

Marks, logos and labels may include symbols of the body or those associated with a programme.

The body shall establish rules applying to references to data and information in an environmental information statement that were validated or verified.

NOTE References to validation/verification opinions, reports and use of marks are further explained in [Annex B](#).

10.3.3 The body's agreement shall require the client to ensure that any opinions or reports of factual findings made public by the client are communicated in their entirety.

10.4 Confidentiality

ISO/IEC 17029:2019, 10.4, shall be followed.

11 Management system requirements

11.1 General

ISO/IEC 17029:2019, 11.1, shall be followed.

11.2 Management review

ISO/IEC 17029:2019, 11.2, shall be followed.

The management review shall be conducted at least once a year, not exceeding 15 months between management reviews.

11.3 Internal audits

ISO/IEC 17029:2019, 11.3, shall be followed.

The internal audit shall be conducted at least once a year, not exceeding 15 months between audits.

11.4 Corrective action

ISO/IEC 17029:2019, 11.4, shall be followed.

11.5 Actions to address risks and opportunities

ISO/IEC 17029:2019, 11.5, shall be followed.

11.6 Documented information

ISO/IEC 17029:2019, 11.6, shall be followed.

Annex A (informative)

Types of opinions

If an opinion is issued, the body shall select an opinion type from a column in [Table A.1](#).

NOTE In case of a mixed engagement (see ISO 14064-3:2019, Annex D), the body can issue more than one opinion or issue opinion(s) and a report of factual findings. [Table A.1](#) provides different terms used to describe opinion types for different programmes.

Table A.1 — Validation/verification opinion name variants

Programme A	Programme B	Programme C	Programme D	Programme E
Unmodified	Unqualified	Positive	Satisfactory	Positive
Modified	Qualified	Qualified positive	Satisfactory with comments	
Adverse	Adverse	Adverse	Unsatisfactory	Negative

NOTE Source: ISO 14064-3:2019, 9.2, Table 1.

NOTE 2 The word “negative” in this context does not refer to a limited level of assurance opinion.

NOTE 3 If there is insufficient evidence to support a conclusion, the body can disclaim the issuance of an opinion.

[Table A.2](#) provides an example of the possible relationship between the type and extent of misstatements and the type of opinion.

Table A.2 — Misstatements and opinion types

Type of misstatement	Extent of misstatement	Opinion type
There is no misstatement	None	Unmodified
The misstatement is not material	Not pervasive	Unmodified/Modified
The misstatement is material	Not pervasive	Modified
	Pervasive	Adverse
There is a misstatement, but the type is unknown	Not pervasive	Modified
	Pervasive	Disclaimed

NOTE 1 When misstatement is not material and not pervasive, opinions may be modified when programme requirements dictate.

NOTE 2 Pervasive misstatements, individually or in aggregate, are those that are:

- not confined to specific elements, classifications or line items of the environmental information statement;
- even if confined, represent a substantial portion of the environmental information statement;
- fundamental to the intended user’s understanding of the environmental information statement.

Annex B (informative)

Reference to validated/verified statements and use of marks

B.1 General

Validation and verification of environmental information statements can result in responsible parties making reference to the findings, conclusions, reports and opinions expressed by validation/verification bodies. Verification or validation bodies have the responsibility (see 10.3) to establish rules governing references made by responsible parties to validation/verification and governing the use of marks.

B.2 References to validated or verified statements

B.2.1 General

In its rules, the body should distinguish between “short-form” and “long-form” references to validated or verified environmental information statements. The body should require that any use of a short-form reference include or make reference to a long-form reference.

NOTE “Include” means that the long-form reference is provided in proximity to the short-form reference in the same medium. “Make reference to” means that a reference to the location of the long-form reference in another medium (e.g. website) is provided in proximity to the short-form reference.

Acceptable references for validated or verified environmental information statements are provided in [Table B.1](#)

Table B.1 — Acceptable references for validated or verified environmental information statements

Subject matter is	Short form	Long form
Historical in nature ^a	“Verified at the reasonable level of assurance”	“In its opinion dated 20xx-xx-xx, [name of body] ^b concluded with reasonable assurance that the data and information in our statement were fairly stated.”
Historical in nature ^a	“Verified at the limited level of assurance”	“In its opinion dated 20xx-xx-xx, [name of body] ^b found no evidence to indicate that the data and information in our statement were not fairly stated.”
Projected or forecast	“Validated”	“In its opinion dated 20xx-xx-xx, [name of body] ^b stated that it had not found any evidence to indicate that the assumptions, methods and limitations that we cited in our statement did not provide a reasonable basis for our projections or forecasts.”
^a Historical data and information submitted for verification may be monitored, estimated or modelled.		
^b When a responsible party refers to a statement as “verified”, the long-form reference applies to any reference implying verification, e.g. by using words such as “verified”, “third-party verified” or “verified by [name of body].”		

B.2.2 References to verification on product statements based on life cycle assessment of products

Verification bodies should apply separate rules for references to mixed engagement of verification and AUP for statements that are based on the life cycle assessment of products. References are provided in [Table B.2](#).

Table B.2 — References to verification and AUP for statements based on the life cycle assessment of products

Type of reference	Short form	Long form
Functional or declared units (mixed engagement)	“Confirmed”	“The upstream and the core data and information in our statement were verified and the downstream data and information were tested by AUP by [name of body] ^a , which did not find any evidence to indicate that our statement was not fairly stated. The verification opinion of the [name of body] ^a and the report of factual findings were issued on 20xx-xx-xx.”
^a When a responsible party refers to subject matter as “verified”, the long-form reference applies to any reference implying verification, e.g. by using words such as “verifier”, “third-party verifier” or “[name of body]”.		

B.3 Use of marks

The rules governing the use of marks should include the use of marks designed to ensure that the responsible party does not use the mark to imply that statements not subject to validation or verification have been validated or verified. For example, use of a body’s mark may include affixing it in a responsible party’s environmental information report next to a description of verified or validated environmental information. It is not permitted to use the body’s mark on environmental information statements which contain information that has not been validated or verified. Examples of the acceptable and unacceptable use of marks are illustrated in [Table B.3](#).

Table B.3 — Examples of acceptable and unacceptable use of marks

Example of an acceptable use of a mark	
[Name of body’s mark]	“Our inventory of greenhouse gas data and information was verified by [name of body].” “In its opinion dated 20xx-xx-xx, [name of body] concluded [with reasonable assurance] that the data and information in our statement were fairly stated.”
Example of an unacceptable use of a mark	
[Name of body’s mark]	“Our inventory of greenhouse gas data and information demonstrated that [responsible party name] had achieved its sustainability goals and had realized science-based targets that put us on a path to transitioning to a low carbon economy in alignment with the objectives of the Paris Agreement.”

Annex C (informative)

Examples of reports of factual findings

C.1 General

This annex provides examples of reports of factual findings that can be issued as the outcome of an AUP engagement. Reports of factual findings are issued as the result of an AUP engagement because issuing an opinion (and thereby providing assurance) is not an option with this type of engagement.

A report of factual findings should include the scope of the subject matter addressed by the procedures, the criteria used for performing them (e.g. ISO 14064-3:2019, Annex C), and the purpose and limitations associated with the agreed-upon verification activities. A report of factual findings should include a statement that the report is to be used solely by the intended user(s) that has(ve) agreed upon the procedures.

The body should report all findings from the application of AUP.

The body should state in its report of factual findings that had it performed additional evidence-gathering procedures or performed a validation or verification of the environmental information, other matters might have come to its attention that would have been reported.

Verifiers who use this engagement type should ensure that the elements given in [Table C.1](#) are included in each report of factual findings.

Table C.1 — Example of an agreed-upon procedures report content

Title	Title that includes the word “impartial”. Addressee (ordinarily the intended user(s)).
Content and roles	Identification of the responsible party. Identification of the subject. A statement that the subject is the responsibility of the responsible party. A statement that the sufficiency of the procedures is solely the responsibility of the intended user(s). A disclaimer of the body’s responsibility for the sufficiency of those procedures.
Methodology	A statement that the procedures performed were those agreed to between the responsible party and the validation/verification body. A statement that the AUP was performed in accordance with [insert standard/programme]. A statement that the report of factual findings is to be used solely by the intended user(s) who have agreed upon the procedures.
Procedures and results	Identification of the purpose for which the AUP were performed. A listing of the specific procedures performed. Where applicable, a description of any agreed-upon materiality limits. A description of the body’s factual findings including sufficient details of errors and exceptions found.
Caveats to the methodology	Where applicable, reservations or restrictions concerning procedures or findings. Where applicable, a description of the nature of the assistance provided by a specialist.

Table C.1 (continued)

Caveats to the AUP report	A statement that the activities performed are a particular type of verification activity that does not result in the issuance of an opinion and does not provide assurance. A statement that had the body performed additional activities, a validation or verification, other matters might have come to light that would have been reported. A statement that the report of factual findings is designed for the intended user and may not be suitable for any other purposes.
Body	Date of the report. Body's address. Body's signature.

C.2 Example of a report of factual findings for a company destroying ozone-depleting substances

The intended user in this example is a company that sought a report on its performance with respect to the operating parameters described in the Code of Good Housekeeping approved by the Montreal Protocol^[19] for facilities destroying ozone-depleting substances (ODS).

Impartial report of factual findings

To the management of the [Destruction Company (hereafter "DC")], Country A and the Ministry of Environment (hereafter "MoE"), Country A.

Agreed-upon procedures

We have performed the evidence-gathering procedures agreed with DC and MoE, Country A's regulatory authority for the destruction of ozone-depleting substances, and enumerated below with respect to the operating parameters described in the Code of Good Housekeeping approved by the Technology and Economic Assessment Panel of the Montreal Protocol (2006, 7th Edition) for facilities destroying ozone-depleting substances for the period 1 August 2017 to 31 October 2017.

Our engagement was undertaken in accordance with the International Organization for Standardization's "Specification with guidance for the validation and verification of greenhouse gas statements" (ISO 14064-3:2019). The sufficiency of these procedures is solely the responsibility of the intended users specified in this report. Consequently, we make no representation regarding the sufficiency of the procedures described below either for the purpose for which this report has been requested or for any other purpose. These procedures were performed solely to assist DC's ODS Destruction Department in evaluating the results of its application of the Code of Good Housekeeping during the period 1 August 2017 to 31 October 2017.

We were not engaged to and did not conduct an assurance engagement, the objective of which would have been the expression of an opinion on DC's statement of conformity to operating parameters described in specified criteria. Accordingly, we do not express such an opinion. Had we performed additional procedures, other matters might have come to our attention that would have been reported to you.

DC is solely responsible for the information that it provided to us for the purposes of this engagement.

Our procedures can be summarized as follows:

1. We checked the applicability of air and water regulations to DC's facility by obtaining communications provided to DC from MoE, the federal environmental regulatory authority.
2. We obtained and checked all records of communications with suppliers during the period August to October 2017 and compared the packaging and shipping instructions to the packaging and containment requirements in DC's ODS acquisition procedures (SOP 12345) and purchase work instruction (WI 12345-002) pg. 4 that states that DC "...will provide packaging and shipping instructions to ODS suppliers".
3. We obtained and checked the weekly tank detection tests for the period August to October 2017 and compared the test procedure to DC's "Recovery Procedures for Destruction Material" (SOP 56789) pg. 7.
4. We obtained and checked the weekly logs for arriving ODS cylinders to confirm conformity to the "Receiving Procedures Upon Arrival of Product" work instruction (WI 56789-001) pg. 5 that requires that the cylinders are checked for leakage upon arrival.
5. We visited DC's ODS storage room to observe the cylinder weighing procedures to confirm conformity to DC's ODS cylinder weighing procedure (SOP 12456).
6. We obtained 12 records of full and empty weights for ODS cylinders and subtracted the empty weight from the full weight for each cylinder to determine whether there was a positive value greater than 5 kg.
7. We obtained DC's ODS sampling and analysis quality control procedures and compared them against "Sampling of Feed Tanks" SOP 34567 p. 8.
8. We obtained and compared the Air Compliance Laboratory of Country A's report on the destruction efficiency of DC's argon arc plasma destruction technology against the minimum acceptable destruction efficiency of 99,99 % with concentrations of pollutants in stack gases and stack gas flow rates expressed on the basis of dry gas at normal conditions of 0°C and 101,3 kPa, and with the stack gas corrected to 11 % O₂.

We report our findings below:

- a. With respect to item 1, a communication from Official #1 of MoE dated 14 January 2016 stated that no air emission or water discharge regulations applied to DC's facility.
- b. With respect to item 2, all (five) records were found to have packing and shipping instructions that conformed to DC's ODS acquisition procedure (SOP 12345) and purchase work instruction (WI 12345-002).
- c. We found that the tank detection tests for the period August to October 2017 conformed to DC's "Recovery Procedures for Destruction Material" (SOP 56789) pg. 7.
- d. We found that the weekly logs for arriving ODS cylinders conformed to the "Receiving Procedures Upon Arrival of Product" work instruction (WI 56789-001) pg. 5.
- e. The observed weighing procedure conformed to DC's ODS cylinder weighing procedure (SOP 12456).
- f. Twelve records had a difference in weight of greater than 5 kg.
- g. DC's quality control procedures conformed to "Sampling of Feed Tanks" SOP 34567 p. 8.
- h. The destruction efficiency reported exceeded minimum requirements.

Because the above procedures do not constitute either a validation or verification in accordance with the International Organization for Standardization’s “Specification with guidance for the validation and verification of greenhouse gas statements” (ISO 14064-3:2019), we do not express any assurance on the statements made to us (dated 25 January 2019) by DC with respect to its conformity to the referenced criteria.

Had we performed additional evidence-gathering procedures, or had we performed a verification of emission reductions obtained from the destruction of ODS in accordance with an applicable greenhouse gas mitigation protocol, other matters might have come to our attention that would have been reported.

Our report is solely for DC’s ODS Destruction Department and for the MoE and may not be suitable for any other purposes.

12 March 2019

MNO’s Verifiers

C.3 Example of a report of factual findings for an issuer of green bonds

The subject matter of the AUP in this example relates to pre-issuance plans of the issuer of a green bond.

Impartial report of factual findings
To the management of The Treasury, Country B (hereafter “The Treasury”) and ABC Bank, Country B (hereafter “ABC Bank”). Agreed-upon procedures We have performed the evidence-gathering procedures agreed with The Treasury and ABC Bank, and enumerated below with respect to the prospective issuance of a green bond for wastewater treatment infrastructure improvements in City Z, Country B. Our engagement was undertaken in accordance with requirements in ISO 14065:2020 and ISO 14030-3:—. The sufficiency of these procedures is solely the responsibility of the intended users specified in this report. Consequently, we make no representation regarding the sufficiency of the procedures described below either for the purpose for which this report has been requested or for any other purpose. These procedures were performed solely to assist The Treasury and ABC Bank in evaluating the results of the application of ISO 14030-1:— and ISO 14030-3:— to the planned issuance of a green bond for wastewater treatment infrastructure improvements in City Z, Country B. We were not engaged to and did not conduct an assurance engagement, the objective of which would have been the expression of an opinion on the conformity of The Treasury’s designation of its City Z 2019 water infrastructure debt obligation as a green bond meeting the requirements of ISO 14030-1:— and ISO 14030-3:—. Accordingly, we do not express such an opinion. Had we performed additional procedures, other matters might have come to our attention that would have been reported to you. The Treasury is solely responsible for the information that it provided to us for the purposes of this engagement.

Our procedures can be summarized as follows:

1. We obtained and checked the final pre-issuance Bond Disclosure Documentation dated 2019-02-15 to determine whether it included an eligibility assessment process for nominated projects.
2. We obtained and checked the final pre-issuance Bond Disclosure Documentation dated 2019-02-15 to determine whether this process met the requirements of ISO 14030-1.
3. We obtained and checked the addition of the planned net proceeds of the bond and compared the sum to the issuer's debt obligation or fair market value (whichever is less) of nominated projects and assets.
4. We reviewed the water infrastructure project type that is intended to be funded by The Treasury's planned debt issuance for City Z and compared this type to the taxonomy of eligible water infrastructure project types in ISO 14030-3.
5. We obtained records used for tracking the proceeds to determine whether an uninterrupted audit trail could be established between intake and disbursement.
6. We obtained and checked records of assets not yet purchased and compared their estimated costs to the amounts of unallocated funds held in earmarked accounts.

We report our findings below:

- a. The final pre-issuance Bond Disclosure Documentation dated 2019-02-15 described an eligibility assessment process.
- b. The final pre-issuance Bond Disclosure Documentation dated 2019-02-15 conformed to the requirements of ISO 14030-1.
- c. The fair market value of nominated projects and assets exceeded the planned net proceeds of the bond.
- d. The water infrastructure project type planned for City Z conformed to one of the types included in the taxonomy of eligible project types in ISO 14030-3.
- e. A continuous audit trail was found for the intake to the disbursement of proceeds.
- f. The amount of funds held in earmarked accounts exceeded by three per cent the estimated costs of assets not yet purchased.

Because the above procedures do not constitute a verification in accordance with the International Organization for Standardization's "Requirements for bodies validating and verifying environmental information" (ISO 14065:2020) and with "Green debt instruments— Part 4: Verification programme requirements" (ISO 14030-4:—), we do not express any assurance on the statements made to us (dated 15 February 2019) by The Treasury, Country B, and ABC Bank, Country B, with respect to its conformity to the referenced criteria.

Had we performed additional evidence-gathering procedures or had we performed an assurance engagement with respect to the conformity of The Treasury's designated green bond for water infrastructure improvements in City Z, other matters might have come to our attention that would have been reported.

Our report is solely for The Treasury, Country B, and ABC Bank, Country B, and may not be suitable for any other purposes.

12 March 2019

MNO's Verifiers

Annex D **(normative)**

Additional requirements applicable to green bond validation, verification and AUP

D.1 General

This annex provides requirements for bodies that perform validation, verification or AUP when the environmental information statement relates to a green bond or green loan. It contains specific requirements related to competencies and processes.

D.2 Personnel competence

The body shall apply the requirements of this document and ISO 14030-4 when forming verification teams or validation teams.

D.3 Process requirements

When fulfilling the process requirements in [Clause 9](#) for statements claiming conformity to the requirements of ISO 14030-1 or ISO 14030-2, the body may perform its engagements in accordance with:

- a) the requirements of ISO 14030-4 and, with the necessary adaptations made, with ISO 14064-3; or
- b) the requirements of ISAE 3000 and, as applicable, ISRS 4400.

Annex E (normative)

Additional requirements applicable to greenhouse gas validation, verification and AUP

E.1 General

This annex provides requirements for bodies that perform validation or verification, or AUP, when the environmental information statement relates to greenhouse gas. It contains specific requirements related to competencies and processes.

E.2 Competencies

E.2.1 General

In addition to the requirements in [Clause 7](#) and ISO 14066, the requirements in [E.2.2](#) to [E.2.3](#) shall apply.

E.2.2 Competencies of personnel

The validation or verification body shall:

- a) ensure that validators and verifiers, and, where required, technical experts, have access to up-to-date information on, and have demonstrated ability to apply skills and knowledge of, greenhouse gas validation or verification processes, requirements, methodologies, activities, other relevant greenhouse gas programme provisions, and applicable legal requirements;
- b) identify training needs and provide, as necessary, training on greenhouse gas validation or verification processes, requirements, methodologies, activities, and other relevant greenhouse gas programme requirements.

E.2.3 Deployment of teams

E.2.3.1 Validation/verification team knowledge

The validation/verification team shall have detailed knowledge of the applicable greenhouse gas programme, including its:

- a) eligibility requirements;
- b) implementation in different jurisdictions as applicable;
- c) validation or verification requirements and guidelines.

E.2.3.2 Validation/verification team technical expertise

The validation/verification team shall have sufficient technical expertise to evaluate:

- a) the greenhouse gas project's, organization's or product's specific greenhouse gas activity and technology;
- b) identification and selection of greenhouse gas sources, sinks or reservoirs;
- c) quantification, monitoring and reporting, including relevant technical and sector issues;

- d) situations that can affect the materiality of the greenhouse gas statement, including typical and atypical operating conditions.

The validation/verification team shall have expertise to evaluate the implications of financial, operational, contractual or other agreements that can affect the greenhouse gas project, organization or product boundaries, including any legal requirements related to the greenhouse gas statement.

E.2.3.3 Validation/verification team data and information auditing expertise

In addition to the requirements given in [7.3.7](#), the following requirements apply.

The validation/verification team shall have data and information auditing expertise to evaluate the greenhouse gas statement of the greenhouse gas project, organization or product, including the ability:

- a) to evaluate the greenhouse gas information system to determine whether the project proponent or organization has effectively identified, collected, analysed and reported on the data necessary to establish a credible greenhouse gas statement, and has systematically taken corrective actions to address any nonconformities related to requirements of the relevant greenhouse gas programme or standards;
- b) to evaluate the impact of the various streams of data on the materiality of the greenhouse gas statement.

E.2.3.4 Specific greenhouse gas project validation team competencies

In addition to the requirements given in [E.2.3.1](#), [E.2.3.2](#) and [E.2.3.3](#), the validation team shall have the expertise to assess processes, procedures and methodologies used:

- a) to select, justify and quantify the baseline scenario, including underlying assumptions;
- b) to determine the conservativeness of the baseline scenario;
- c) to define the baseline scenario and greenhouse gas project boundaries;
- d) to demonstrate equivalence between the type and level of activities, goods or services of the baseline scenario, and the greenhouse gas project;
- e) to demonstrate that greenhouse gas project activities are additional to baseline scenario activities;
- f) to demonstrate conformity, if appropriate, to greenhouse gas programme requirements such as secondary effects (leakage) and permanence.

NOTE ISO 14064-2 includes requirements and guidance on the principle of conservativeness and the concept of equivalence.

In addition to the requirements given in [E.2.3.1](#), [E.2.3.2](#) and [E.2.3.3](#), the validation team shall have knowledge of relevant sector trends that can have an impact on the selection of the baseline scenario.

E.2.3.5 Specific greenhouse gas project verification team competencies

In addition to the requirements given in [E.2.3.1](#), [E.2.3.2](#) and [E.2.3.3](#), the project verification team shall have the expertise appropriate to assess processes, procedures or methodologies used:

- a) to evaluate consistency between the validated greenhouse gas project plan and the greenhouse gas project implementation;
- b) to confirm the ongoing appropriateness of the validated greenhouse gas project plan, including its baseline scenario and underlying assumptions.

E.2.3.6 Specific greenhouse gas product verification team competencies

In addition to the requirements given in [E.2.3.1](#), [E.2.3.2](#) and [E.2.3.3](#), the product verification team shall have competence on:

- life cycle assessment methodology;
- the product category rule (PCR) or product category rule for carbon footprints (CFP-PCR) applicable to the specific verification;
- the structure of the database applicable on the specific verification.

E.3 Process requirements

E.3.1 General

When fulfilling the process requirements in [Clause 9](#) for greenhouse gas opinions, the body shall perform its engagements in accordance with the requirements of ISO 14064-3.

E.3.2 Specific conditions for the carbon footprint of products (CFP) systematic approach

When verifying a CFP that is formulated in accordance with a CFP systematic approach (see ISO 14067:2018, Annex C), the verification body shall confirm the ongoing effectiveness of the CFP system.

E.4 Outsourcing

In the absence of greenhouse gas programme prohibitions on outsourcing, the validation or verification body may outsource activities (see [7.4](#)) but shall require the outsourced body to provide independent evidence that demonstrates its conformity to this document and with ISO 14064-3.

Additional requirements applicable to validation, verification and AUP associated with reporting investments and financing activities related to climate change

This annex provides requirements for bodies that perform validation or verification, or AUP, when the environmental information statement relates to reporting investments and financing activities related to climate change. It contains specific requirements related to competencies and processes.

The body shall apply the requirements of this document and ISO 14097 when forming verification teams or validation teams.

When fulfilling the process requirements in [Clause 9](#) for statements claiming conformity to the requirements of ISO 14097, the body may perform its engagements in accordance with:

- a) the requirements of ISO 14097:—, Clause 9, and, with the necessary adaptations made, with ISO 14064-3; or
- b) the requirements of ISAE 3000 and, as applicable, ISRS 4400.

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- [3] ISO 14016, *Environmental management — Guidelines on the assurance of environmental reports*
- [4] ISO 14030-1:—,³⁾ *Environmental performance evaluation — Green debt instruments — Part 1: Process for green bonds*
- [5] ISO 14030-3:—,⁴⁾ *Environmental performance evaluation — Green debt instruments — Part 3: Taxonomy*
- [6] ISO 14033, *Environmental management — Quantitative environmental information — Guidelines and examples*
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- [8] ISO 14046, *Environmental management — Water footprint — Principles, requirements and guidelines*
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- [11] ISO 14064-2, *Greenhouse gases — Part 2: Specification with guidance at the project level for quantification, monitoring and reporting of greenhouse gas emission reductions or removal enhancements*
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- [18] ISRS 4400, *Engagements to Perform Agreed-Upon Procedures Regarding Financial Information*
- [19] UNEP. *Handbook for the Montreal Protocol on Substances that Deplete the Ozone Layer*. Seventh Edition. United Nations Environment Programme (UNEP), 2006

3) Under preparation. Stage at the time of publication: ISO/DIS 14030-1:2020.

4) Under preparation. Stage at the time of publication: ISO/DIS 14030-3:2020.

